THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a^2 - b^2 = 225$ and $a - b = 9$, then $a + b$ equals: 1

(a) 25 **(b)** 26 **(c)** 9 **(d)** 136

2. The value of $27^2 - 24 \times 30$ is: 1

(a) 27 **(b)** 9 **(c)** 0 **(d)** 6

3. The sum of three consecutive integers is 102. The largest of them is: 1

(a) 36 **(b)** 33 **(c)** 34 **(d)** 35

4. A three-digit number with digits a, b, c is reversed. If $a - c = 3$, the difference between the number and its reverse is: 1

(a) 297 **(b)** 27 **(c)** 270 **(d)** 300

5. A two-digit number exceeds the number formed by reversing its digits by 45. The difference of its digits is: 1

(a) 5 **(b)** 45 **(c)** 4 **(d)** 6

6. If $x + \frac{1}{x} = 5$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 23 **(b)** 10 **(c)** 27 **(d)** 25

7. The sum of the dates in a 2×2 calendar block is 88. The smallest date in the block is: 1

(a) 17 **(b)** 19 **(c)** 18 **(d)** 26

8. If $a^2 - b^2 = 360$ and $a - b = 12$, then $a + b$ equals: 1

(a) 31 **(b)** 12 **(c)** 189 **(d)** 30

9. **Assertion (A):** The sum of three consecutive numbers n, $n + 1$, $n + 2$ is $3(n + 1)$. 1

Reason (R): The sum of three consecutive numbers is always divisible by 6.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $3x + 4x = 7x^2$. 1

Reason (R): Like terms can be added by adding their coefficients.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $x + y = 23$ and $x - y = 7$, find $x^2 - y^2$. 2

12. Prove that $(n+3)^2 - (n-3)^2 = 12n$ and use it to find the value of $(8+3)^2 - (8-3)^2$. 2

13. The sum of the digits of a two-digit number is 13 and the tens digit exceeds the ones digit by 1. Find the number. 2

14. If $x + y = 31$ and $x - y = 4$, find $x^2 - y^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The sum of a two-digit number and its reverse is 154 and the digits differ by 2. Find the number (tens digit larger). 3

16. If $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

17. The sum of the dates of a 2×2 block in a calendar is 68. Find the four dates and show that the block's sum is always $4\times$(smallest date)$+16$. 3

OR

The sum of a two-digit number and its reverse is 88 and the digits differ by 2. Find the number (tens digit larger). 3

18. If $x + \frac{1}{x} = 6$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

OR

The sum of the dates of a 2×2 block in a calendar is 80. Find the four dates and show that the block's sum is always $4\times$(smallest date)$+16$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ishita arranges matchsticks to make a row of squares: 1 square needs 4 sticks, 2 squares need 7 sticks and 3 squares need 10 sticks, and so on. 5

(i) How many sticks are needed for 4 squares?

(ii) Write an expression for the number of sticks for n squares.

(iii) How many sticks are needed for 20 squares?

(iv) How many squares can be made with 100 sticks?

20. Prove that $(5n+2)^2 - (5n-2)^2 = 40n$ and use it to evaluate it for $n = 3$. 5

OR

A three-digit number has digits a, b, c with $a - c = 4$. Show that the number minus its reverse is divisible by 99 and find the difference. 5

END OF QUESTION PAPER — Total: 40 Marks